Homework assignment for 02YELMA
Chapter: Electrostatics

1. Four equal charges, $+q$, are located at the corners of a square with side length a. What is the net electric field in the center of the square? If we replace one of the charges with $-q$, what would be the electric field and what would be the net force on a test charge Q in the center?

2. Find the electric field a distance z above the center of a circular loop of radius r, which carries a uniform line charge λ.

3. Find the electric field inside and outside of a charged sphere with charge density $\rho = kr$, where k is a constant and r is the distance from the center of the sphere.

4. Depending on the result of Q3, calculate the potential inside and outside of the sphere.

5. If you consider the situation in Q1 again (four equal charges, $+q$, are situated at the corners of a square), how would the field lines look?

6. Consider a uniform disc-shaped surface charge σ with radius R. Find the potential at a distance z above its center. Calculate the electric field using the potential. (Hint: Consider the substitution $u = r^2 + z^2$)

7. Consider a sphere of radius R with a static, spherically symmetric volume charge density given by
$$\rho(r) = \rho_0 \left(1 - \frac{r}{R}\right), \qquad 0 \le r \le R,$$
and $\rho(r) = 0$ for $r > R$.

   (a) Determine the total charge Q of the sphere in terms of ρ_0 and R.

   (b) Using Gauss's law, find the electric field $E(r)$:

      i. for $r < R$,

      ii. for $r > R$.

   (c) Using the field-energy expression
   $$U = \frac{\varepsilon_0}{2} \int_{\text{all space}} E^2 \, d\tau,$$
   compute the total electrostatic self-energy of the charge distribution.

   (d) Express your final answer in terms of Q and R only.

   (e) Compare your result with the well-known result for a uniformly charged sphere:
   $$U_{\text{uniform}} = \frac{3Q^2}{20\pi\varepsilon_0 R}.$$

8. Three charges are to be assembled from infinity:

   - Two charges $+q$
   - One charge $-2q$

   They are arranged at the vertices of an equilateral triangle of side length a.

   (a) Compute the total formation energy of the configuration.

   (b) Determine whether the configuration is energetically favorable (negative total energy).

   (c) If the charge $-2q$ is instead placed at the center of the triangle (distance $a/\sqrt{3}$ from each vertex), recompute the formation energy.

   (d) Which configuration is energetically preferred?

   Assume vacuum permittivity ε_0 and neglect self-energies.

9. Show explicitly what is the energy required to assemble a point charge by using the formula for the energy of a given electric field. Interpret your result.

10. If we assume that the electron has a uniform surface charge distribution in the form of a spherical shell, and we calculate its rest energy using the formula $E = mc^2$, what would be its radius *in meters*? Find relevant quantities on the Internet such as m, c, etc., ensuring that they are compatible in terms of units.

11. A metal sphere of radius R carries a total charge Q. What is the force of repulsion between the northern and southern hemispheres?

12. Suppose that the plates of a parallel plate capacitor move closer together by an infinitesimal distance ϵ, as a result of their mutual interaction. What is the amount of work done by electrostatic forces, in terms of the electric field E and the area of the plates A? (Hint: Use $P = \frac{\epsilon_0}{2}E^2$)

13. Find the capacitance *per unit length* of two cylindrical coaxial metal tubes, of radii a and b.

14. Four particles are placed in the following positions: $-2q \to (0, -a, 0)$, $-2q \to (0, a, 0)$, $q \to (0, 0, -a)$, $3q \to (0, 0, a)$. Find a simple approximate formula for the potential that is valid at points far from the origin.

15. Two charges are placed in two different arrangements: (a) $3q$ at $(0, 0, a)$, and $-q$ at $(0, 0, 0)$, and (b) $3q$ at $(0, 0, 0)$, and $-q$ at $(0, 0, -a)$. For each arrangement, determine the monopole moment (in other words, the monopole contribution), the dipole moment, and the approximate potential (in spherical coordinates) using these moments.

16. A pure dipole p is located at the origin, oriented in the $-z$ direction. What is the force experienced by a point charge located at the coordinates $(a, 0, 0)$?

17. Two pure electric dipoles $\vec{p}_1$ and $\vec{p}_2$ are a distance r apart along the x-axis. What is the torque experienced by $\vec{p}_2$ due to the presence of $\vec{p}_1$? ($\vec{p}_1$ and $\vec{p}_2$ are oriented along the $\hat{z}$ and $\hat{x}$ directions, respectively.)

18. A sphere of radius R has a polarization $\vec{P}(\vec{r}) = k\vec{r}$ where k is a constant and $\vec{r}$ is the vector from the center. Calculate the bound charges σ_b and ρ_b, and find the field inside and outside the sphere.

Given for 13 and 14: The electric field created by a pure dipole:

$$\vec{E}_{dip}(r, \theta) = \frac{p}{4\pi\epsilon_0 r^3}\left[2\cos(\theta)\hat{r} + \sin(\theta)\hat{\theta}\right]$$